Classical molecular dynamics study for defect sink behavior in oxide dispersed strengthened alloys

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***Abstract*— Oxide Dispersed Strengthened (ODS) alloys are considered as a potential candidate for future generation reactors and can even be graded as better radiation resistant materials than the ones already in use. Molecular dynamics (MD) simulation study for defect sink properties of ODS alloys have been investigated at 700K using LAMMPS (Atomic/Molecular massively parallel simulator software. SRIM (stopping range of ions in matter) code was used to see the approximate range of the ions and damage cascade profile. It has been observed that the oxide particle attracts vacancies/interstitials produced in the base matrix near the interface. The modeled structure looked amorphous near the interface but appears crystalline in the rest region. The modeled structure has defect sink property near the interface. The area for vacancy/interstitial cluster shrinks with the time evolution. Initially, the defects are found scattered near the interface but with time evolution, these recombine and annihilate. After the bombardment Fe atoms show better recover than the oxides, thus it could be concluded that Fe has more tendency to recover than oxide. This atomistic investigation would be beneficial to analyze the multiscale behavior determination of the materials being experimented under severe conditions. This defect sink behavior study would help to assess better radiation resistant materials for the current and future generation reactors.**

Keywords*— Oxide dispersed strengthened alloy, Nano oxide, stability molecular dynamics simulation, clusters

I. Introduction

In the recent extensive research on the advanced fission and fusion reactors, structural and cladding materials have been found crucial for efficiency and reliability of reactors[1-4]. The overall performance of the reactors depends largely on the characteristics of the materials used. As these material are adversely effected by harsh radiation environments including ;high temperature, stress and chemical changes during irradiation interactions[5-7].Keeping in view the harsh operating environment of the reactor, factors like corrosion resistance, ease of workability and thermal conductivity should also be considered in addition to the general mechanical strength of a material. Another significant requirement for the structural materials is to have radiation stability under high neutron radiation and low neutron absorption cross-section [3, 8-10]. The aforementioned conditions restrict the future use of certain materials in the nuclear industry. Therefore the international forum for fission and fusion reactors stressed on the need to explore new materials in order to meet the demands of nuclear industry[11-12]. Lately, it is has been observed that ferritic/martensitic (F/M) stainless steel is one of the promising candidates for fast reactors cladding materials. Since they have the characteristic of large thermal conductivity and small expansion coefficient. At 300-550℃ it exhibits good thermal conductivity and radiation resistant behavior. But it shows drastic results such as radiation embrittlement in both cases, if the temperature is decreased than 300℃ or if it is increased above 550℃[11,13-16]. Keeping in view it vulnerability under high and low temperatures, it is not an ideal candidate for the development of the advanced generation reactors as high temperature mechanical properties are required by it. In contrast to that ODS steel working temperature is relatively high with excellent anti-radiation stability as compared to F/M stainless steel.Ordinary ODS steel reinforced phase is achieved by the direct addition of oxides, and nanostructure ODS are also called Nanostructured Ferritic(NFA) alloys steel with Y-Ti-O phase.It is formed through mechanical alloying by direct addition of oxides Y_2O_3 solid solution through precipitation solid solution which form highly dispersed oxides having the size of a few nanometers[1,17]. In order to achieve the best results, it is required to study the behavior of the material at different time scales displaying better radiation resistance. A simulation is an important tool in investigating the behaviors of materials at different time scales[6,15].MD simulation is a tool to simulate the primary defect cascade and interaction of defects with the interface in nanosecond scale[7,10]. This study investigates two aspects firstly damage cascade generation in ODS alloys when bombarded with Y ion at an energy of 10,20 KeV. The estimated energy for the present model was determined by SRIM code[18]. Secondly, it investigates the defect interface behavior of oxide plus recovery of Y/O atoms after the bombardment.

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